

Model Evaluation Summary

This report summarizes the performance of the models trained by the `models/train.py` script.

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Final Model Comparison

The following table shows the performance metrics for the baseline and machine learning models on the test set.

Model	F1 Score	PR-AUC	Precision	Recall
zscore	0.9938	1.0000	0.9877	1.0000
logistic	0.7544	0.9870	1.0000	0.6056
random_forest	0.9984	1.0000	0.9969	1.0000

Conclusion

- **Best Model (by PR-AUC):** The Z-Score and Random Forest models both achieved a near-perfect PR-AUC. Given its simplicity, the **Z-Score model** is identified as the best performing model.
- **All models were successfully logged to MLflow.**
- **The Random Forest model was saved as the production artifact.**

This data is sourced directly from the final, successful execution of the `train.py` script.